UNIVERSITI TEKNOLOGI
MARA
MARKING SCHEME

COURSE	:	DATABASE DESIGN AND DEVELOPMENT
COURSE CODE	:	ICT450
EXAMINATION	:	2026 (MOCK)
TIME	:	3 HOURS

MARKING SCHEME NOTES

1. This marking scheme accompanies the ICT450 mock paper (2026).

2. Five questions x 20 marks = 100 marks.
Questions and answers are for local study use only.

FOR MARKERS' USE ONLY

This marking scheme consists of 6 printed pages

QUESTION 1

Database Concept

a) A database is an organised collection of related data. A DBMS is software that creates, stores, manages and controls access to a database. A database system is the database, DBMS, application programs, people, procedures and hardware working together.

Marking points: database definition (2); DBMS definition (2); database system definition (2)

b) The limitations include data redundancy, data inconsistency, difficult data sharing, weak concurrency control, limited security, program-data dependence, and difficult backup/recovery.

Marking points: four limitations x 2 marks each

c) A DBMS provides controlled data sharing, integrity enforcement, security and authorisation, concurrency control, backup and recovery, and a data dictionary.

Marking points: three functions x 2 marks each

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QUESTION 2

Relational Data Model

a) Primary key: (StudentID, CourseID). Candidate keys: (StudentID, CourseID); StudentID alone is not a candidate key because a student enrols in many courses, and CourseID alone is not a candidate key because a course has many students.

Marking points: primary key (2); candidate-key explanation (2)

b) Entity integrity: a primary-key attribute cannot be null. Referential integrity: a foreign-key value must match an existing primary-key value or be null. Example: ENROL.StudentID must exist in STUDENT.StudentID.

Marking points: entity integrity definition/example (3); referential integrity definition/example (3)

c) Tuple = a row; attribute = a column; degree = number of attributes; cardinality = number of tuples.

Marking points: one mark each

d) StudentName repeats for every enrolment, so a name change may be missed in some rows (update anomaly). Deleting the last enrolment for a course may lose course data (delete anomaly). Inserting a new course before any student enrols is difficult (insert anomaly).

Marking points: two anomalies explained x 3 marks

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QUESTION 3

ERD

a) Entities: MEMBER, BOOK, PUBLISHER, LOAN. Relationships: PUBLISHER publishes BOOK (1:M); MEMBER borrows BOOK through LOAN (M:N resolved by LOAN); LOAN has loan date and due date.

Marking points: entities (2); relationships (3); cardinalities (3)

b) A correct ERD shows MEMBER, BOOK, PUBLISHER and LOAN, with 1:M from PUBLISHER to BOOK, 1:M from MEMBER to LOAN, and 1:M from BOOK to LOAN. LOAN is the associative entity for the M:N borrowing relationship.

Marking points: four entities (2); three relationships/cardinalities (3); associative LOAN entity (1)

c) PUBLISHER(PublisherID PK, PublisherName); BOOK(BookID PK, Title, PublisherID FK); MEMBER(MemberID PK, MemberName); LOAN(LoanID PK, MemberID FK, BookID FK, LoanDate, DueDate).

Marking points: four relations (4); PK/FK marks (2)

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QUESTION 4

Normalization

a) Candidate key: (OrderID, ProductID). FDs: OrderID → CustomerID, CustomerName; CustomerID → CustomerName; ProductID → ProductName; (OrderID, ProductID) → Quantity, UnitPrice.

Marking points: candidate key (2); FD list (2)

b) 1NF: all attributes are atomic. 2NF: 1NF and every non-key attribute is fully functionally dependent on the whole candidate key. 3NF: 2NF and no non-key attribute is transitively dependent on the key.

Marking points: 1NF (2); 2NF (2); 3NF (2)

c) Partial: OrderID → CustomerID/CustomerName and ProductID → ProductName depend on part of the key. Transitive: OrderID → CustomerID → CustomerName.

Marking points: partial dependencies (2); transitive dependency (2)

d) CUSTOMER(CustomerID PK, CustomerName); PRODUCT(ProductID PK, ProductName); ORDER_HEADER(OrderID PK, CustomerID FK); ORDER_LINE(OrderID PK, ProductID PK, Quantity, UnitPrice, FKs to ORDER_HEADER and PRODUCT).

Marking points: CUSTOMER (1.5); PRODUCT (1.5); ORDER_HEADER (1.5); ORDER_LINE (1.5)

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QUESTION 5

Database Design

a) Database initial study, database design, implementation and loading, testing and evaluation, operation, maintenance and evolution. Activities include requirements analysis, conceptual/logical design, creating storage structures, loading data, testing, and tuning.

Marking points: four phases x 2 marks (phase + activity)

b) SDLC covers the whole information-system life cycle; DBLC focuses on the database component within that system and runs through its own design, implementation, testing, operation and maintenance phases.

Marking points: SDLC scope (2); DBLC scope (2)

c) Top-down starts from the overall data model and breaks it into components. Bottom-up starts from individual data elements and combines them into a model.

Marking points: top-down (2); bottom-up (2)

d) Centralized design is controlled by one design team or location. Decentralized design distributes design work across teams or locations and requires reconciliation.

Marking points: centralized (2); decentralized (2)

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